

Finding the Sweet Spot: An Empirical Study on Dataset Size, Performance, and Efficiency in Relation Extraction

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Abstract— Relation extraction (RE) is a critical task in natural language processing that identifies relationships between entities within text. While increasing the size of annotated datasets generally enhances model performance, the cost and effort associated with human annotation can be prohibitive. This study investigates the impact of dataset size on RE performance using the Semantic Evaluation 2010 Task 8 dataset. Four deep neural models—Attention-Based Convolutional Neural Network, Attention-Based Bidirectional Long Short-Term Memory Network, Relation Bidirectional Encoder Representations from Transformers, and Matching the Blanks—were evaluated across eight subsets of the dataset to simulate size reduction. Performance metrics such as accuracy, precision, recall, and F1-score were analyzed. The results indicate that although larger datasets improve model performance, the benefits diminish beyond a certain size, illustrating the law of diminishing returns. These findings emphasize the importance of balancing dataset size and quality for cost-effective training and suggest strategies such as data augmentation and transfer learning to optimize performance with limited data.

Keywords—Relation Extraction, Language Model, Deep Learning, Diminishing Returns

I. INTRODUCTION

The size of a dataset significantly impacts the performance of machine learning models. Larger datasets generally enhance classification accuracy, while smaller datasets are at a greater risk of overfitting [1], [2]. Data collection poses various challenges, including privacy concerns, limited samples for rare cases, and legal restrictions. Furthermore, working with large datasets demands more time and computational resources for training, which may be limited [3]–[5]. Despite ongoing efforts, a clear definition of a small dataset remains elusive. Some frameworks estimate the minimum number of samples required for accurate results, while others draw on algorithmic information theory or establish size ranges based on prior research [6]–[8].

Prior studies on small datasets have established size ranges and highlighted the challenges they pose for machine learning. Small datasets often lack sufficient detail, making it difficult for classification algorithms to detect patterns and increasing the risk of overfitting, even when a validation set is employed [9]. Consequently, classification models trained on small datasets may become unreliable and biased. While efforts have been made to enhance classification accuracy on small datasets, further research is necessary to understand how

dataset size impacts algorithm performance and to develop robust methodologies for trend identification and model development.

Recent efforts to address the challenges of small datasets have focused on enhancing classification model performance and augmenting training sets. Techniques include generating virtual samples, leveraging fuzzy rules, and employing similarity-based algorithms to create or extend dataset attributes. Methods like particle swarm optimization and Gaussian-based data generation have demonstrated improved results, emphasizing smoothness and attribute relationships. Additionally, extending attribute information has been shown to improve learning performance and mitigate data sparsity in small datasets [10]–[16].

Research has primarily focused on improving the accuracy of classification algorithms on small datasets, with studies exploring how dataset size impacts performance across domains such as sentiment classification, object detection, plant disease classification, medical classification, and information retrieval [17]–[21].

This study investigates the effect of dataset size on supervised machine learning models for Relation Extraction (RE), a task that identifies relation tuples from text. Using the Semantic Evaluation (SemEval)-2010 Task 8 dataset [22], we conducted experiments on four deep neural network models, creating eight smaller subsets to simulate size reduction. Model performance metrics were analyzed, including accuracy, precision, recall, and F-score.

The contribution of this paper is evaluating the impact of dataset size on supervised machine learning models for RE, using subsets to simulate size reduction. It compares four deep neural network models—Attention-Based Convolutional Neural Networks (Att-CNN) [23], Attention-Based Bidirectional Long Short-Term Memory (Att-BiLSTM) [24], Relation Bidirectional Encoder Representations from Transformers (RBERT) [25], and Matching the Blanks (MTB) [26]—across various dataset sizes, using metrics such as accuracy, precision, recall, and F-score. Focusing on the SemEval-2010 Task 8 dataset, the study examines how model performance changes with dataset size and identifies challenges in optimizing models with limited data. The findings highlight diminishing returns as dataset size increases.

The structure of this paper is as follows. Section II provides an overview of the classification models used in this

study. Section III details the methodology, including dataset partitioning and sub-dataset selection. Section IV presents the experimental results, analyzing the impact of dataset size on model performance and highlighting key trends, such as diminishing returns as dataset size increases and the importance of dataset size thresholds. Finally, Section V concludes the study, summarizing the findings.

II. CLASSIFICATION MODELS

We applied four different deep neural classification models, including attention-based neural networks and a deep transformer model designed to deal with the relation extraction problem.

A. Relation Classification from Attention-Based Neural Networks

This section describes two attention-based neural networks that extract semantic relations between entities. The attention mechanism is crucial in RE, as not all words equally contribute to the semantic relationship. Identifying key clues that define the relation is essential. We will focus on two models: Att-CNN and Att-BiLSTM networks.

1) Att-CNN: This architecture incorporates an attention mechanism within Convolutional Neural Networks (CNNs) to identify words most relevant to the target entities and merge their representations into a sentence vector. Numerical representations are generated by combining word embeddings, position embeddings, and part-of-speech tag embeddings. The sentence convolution vector module uses a convolutional layer to aggregate local features for global relation prediction. A non-linear function generates a feature map and a max-pooling operation extracts the most essential features. The attention-based context vector module employs an attention mechanism to quantify the contextual importance of words concerning target entities directly. Each word and entity in the sequence is processed through a multi-layer perceptron to compute attention weights.

The concatenation of each entity and word representation generates a new word representation, producing a sentence representation for each entity based on attention scores that highlight informative words. The sentence convolution vector and the attention-based context vectors are then merged into a fixed-length feature vector and passed to the output layer for classification.

2) Att-BiLSTM Networks: This architecture leverages an attention mechanism to identify words with the closest semantic relationship to the target entities, using outputs from a Bidirectional Long Short-Term Memory (BiLSTM) network without external knowledge. Unlike standard Long Short-Term Memory networks, which process sequences in a single direction and ignore future context, BiLSTM enhances performance by incorporating past and future context through hidden-to-hidden links flowing in both time orders. This enables the model to utilize comprehensive sequence information effectively. The model takes a sentence as input, processed in the embedding layer to generate low-dimensional vectors for each word. These vectors are passed through a BiLSTM layer that extracts high-level features by combining forward and backward sequence contexts. The outputs are merged using an element-wise sum operation. An attention layer then assigns weights to word-level features, creating a sentence-level feature vector. Finally, using the

log-likelihood loss function, a softmax classifier predicts the label based on the hidden representation.

B. Relation Classification from Deep Transformer Model

The goal is to classify relation statements in a supervised manner using pre-trained language models. This study employs the R-BERT model, which enhances relation classification by integrating pre-trained Bidirectional Encoder Representations from Transformers (BERT) embeddings with target entity information, and the MTB model, which introduces the MTB objective to refine relation representations beyond standard BERT pre-training.

1) R-BERT: R-BERT enhances RE by incorporating entity location information into the pre-trained BERT model. Unique tokens are added for each entity pair, including this information in the sentence vector representation.

2) MTB for Relation Learning: MTB improves relation representation using a novel MTB objective. The model uses entity marker tokens in BERT inputs and extracts fixed length relation representations based on entity start states. It calculates semantic similarity scores between relation statements, identifying related or unrelated pairs without requiring predefined ontologies or labeled training data. Despite limited training data, MTB demonstrated strong performance on the SemEval-2010 Task 8 dataset.

Table 1: Dividing the SemEval-2010 Task 8 Dataset to smaller subsets.

Dataset Notation	Portion	Number of instances
SE.1	1/128	62
SE.2	1/64	125
SE.3	1/32	250
SE.4	1/16	500
SE.5	1/8	1000
SE.6	1/4	2000
SE.7	1/2	4000
SE.8	1	8000

III. METHODOLOGY

In our research on the influence of dataset size on relation extraction (RE) performance, we employed an experimental approach. We specifically selected the SemEval-2010 Task 8 dataset and divided it into eight distinct groups for analysis. We randomly extracted seven subsets from the original dataset using sampling techniques, as documented in Table 1. The details of the partitioning protocol are outlined in Algorithm 1.

```

Require: SE % SemEval-2010 Task 8 Dataset
Ensure: SE.1, SE.2... SE.8
% Eight Subsets of Original Dataset
% Define subset sizes:
Subset Sizes ← [1/128, 1/64, 1/32, 1/16, 1/8, 1/4, 1/2, 1]
% Set random seed for reproducibility:
Set Random Seed
for i = 1 to 8 do
  X ← Subset Sizes[i]
  % Get size of the subset:
  SE.i ← Uniformly Random Sample of Size X × |SE| from SE
end for
End

```

Algorithm 1: Dataset Partitioning for Incremental Evaluation.

The objective of this study is to investigate how classification performance is influenced by increasing the size of the same dataset. The classifiers' performance is evaluated based on accuracy, precision, recall, and F-score. After preprocessing, four neural network models were trained on all subsets of the original dataset. The following subsections will discuss the selection of the dataset and sub-datasets, the classification models employed, and the performance assessment metrics used. The SemEval-2010

Task 8 dataset is widely recognized for supervised relation extraction (RE). It contains 8K training entity pairs and 2,717 test entity pairs, sourced from web sentences and classified into nine bidirectional relation types. These types were selected to minimize semantic overlap while encompassing a broad range of relations, including abstract categories such as Product-Producer and Content-Container. The dataset was provided as part of a shared challenge on RE, featuring nine relation types and an "Other" category.

Table 2: Performance metrics (Accuracy, Precision, Recall, and F-score) of different Classification models (Att-CNN, Att-BiLSTM, R-BERT, and MTB) evaluated across eight subsets of the dataset (SE.1 to SE.8).

Classification Model	Performance Metric	SE.1	SE.2	SE.3	SE.4	SE.5	SE.6	SE.7	SE.8
Att-CNN	Accuracy	0.22	0.26	0.37	0.48	0.58	0.66	0.73	0.77
	Precision	0.21	0.16	0.26	0.41	0.57	0.64	0.74	0.79
	Recall	0.20	0.25	0.30	0.41	0.59	0.69	0.74	0.79
	F-score	0.16	0.19	0.27	0.40	0.56	0.66	0.74	0.79
Att-BiLSTM	Accuracy	0.30	0.45	0.46	0.60	0.66	0.72	0.75	0.79
	Precision	0.34	0.42	0.42	0.59	0.65	0.70	0.76	0.81
	Recall	0.33	0.39	0.42	0.60	0.73	0.76	0.77	0.80
	F-score	0.25	0.37	0.38	0.57	0.65	0.71	0.75	0.80
R-BERT	Accuracy	0.20	0.26	0.41	0.65	0.73	0.79	0.82	0.84
	Precision	0.24	0.26	0.34	0.65	0.71	0.75	0.77	0.79
	Recall	0.13	0.12	0.28	0.56	0.69	0.77	0.81	0.83
	F-score	0.07	0.11	0.27	0.54	0.69	0.76	0.79	0.81
MTB	Accuracy	0.19	0.22	0.38	0.59	0.71	0.78	0.83	0.85
	Precision	0.19	0.23	0.32	0.58	0.69	0.74	0.76	0.79
	Recall	0.11	0.12	0.25	0.51	0.66	0.73	0.78	0.80
	F-score	0.03	0.09	0.20	0.51	0.67	0.74	0.78	0.80

IV. RESULTS AND DISCUSSION

This section evaluates the performance of four classifiers: Att-CNN, Att-BiLSTM, R-BERT, and MTB, using subsets of the SemEval-2010 Task 8 dataset. These subsets, labeled SE.1 to SE.8, represent increasing fractions of the dataset, from 1/128 to the full size. Metrics including accuracy, precision, recall, and F1-score were analyzed to assess model effectiveness, as summarized in Table 2.

Att-CNN demonstrated consistent improvement as the dataset size increased, with accuracy rising from 0.22 on SE.1 to 0.77 on SE.8. Similarly, precision, recall, and F1-score followed this trend, with the latter improving sharply from 0.16 on SE.1 to 0.79 on SE.8, highlighting the model's reliance on larger datasets for optimal performance. Att-BiLSTM showed the most substantial gains, starting with an accuracy of 0.30 and an F1-score of 0.25 on SE.1 and achieving 0.79 and 0.80, respectively, on SE.8. Notable improvements occurred between SE.3 and SE.5, where dataset size appears critical for this model's performance. R-BERT, initially underperforming, saw rapid improvement as dataset size increased, achieving an accuracy of 0.84 and an F1-score of 0.81 on SE.8, with significant gains between SE.4 and SE.6 due to its transformer-based architecture. MTB followed a similar trajectory, starting with an F1-score of 0.03 on SE.1 and improving steadily to 0.80 on SE.8, relying heavily on larger datasets for effective performance. The results reveal a consistent trend across all classifiers:

substantial performance gains occur as dataset size increases from SE.1 to SE.4, representing the steepest learning curve. Beyond SE.6 (50% of the dataset), improvements plateau, demonstrating the concept of diminishing returns. This trend holds across all metrics and classifiers, regardless of complexity, from simpler models like Att-CNN and MTB to more advanced architectures like R-BERT and Att-BiLSTM. The SE.6 subset marks a practical threshold for balancing performance improvements with computational and annotation costs, emphasizing the critical relationship between data availability and model efficacy.

Despite having more than half of the dataset available beyond the green line in Figures 1-2, none of the models—Att-CNN, Att-BiLSTM, R-BERT, or MTB—show significant performance improvement. This suggests a saturation effect, where additional data no longer contributes meaningfully to model accuracy. This could reflect architectural limitations for the Att-CNN and Att-BiLSTM models, which we need to understand better for future improvements. The flattening curve indicates that the pre-trained BERT models have already captured the key patterns in the data. The plateau we observe implies that, beyond a certain point, factors other than dataset size, such as model complexity or task-specific challenges, become more critical in driving performance improvements.

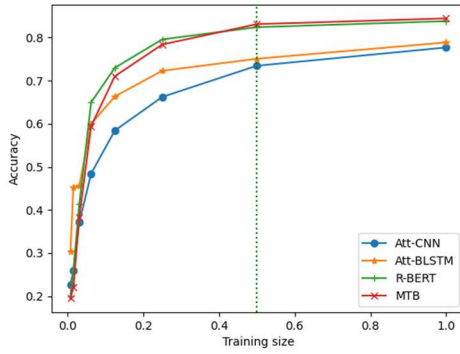


Fig. 2. Impact of training data size on the accuracy of Att-CNN, Att-BiLSTM Networks, and pre-trained BERT models (R-BERT and MTB) used for RE task.

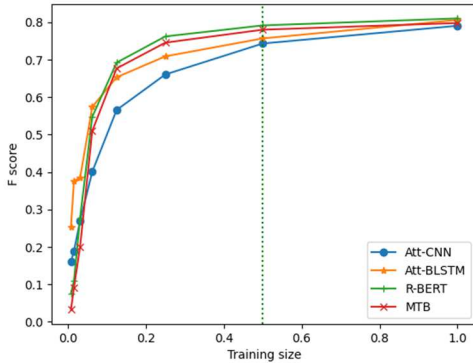


Fig. 1. Impact of training data size on the F-score of Att-CNN, Att-BiLSTM Networks, and pre-trained BERT models (R-BERT and MTB) used for RE task.

V. CONCLUSION

This research offers a comprehensive evaluation of the impact of dataset size on the performance of relation extraction (RE) systems, utilizing four distinct classifiers: Att-CNN, Att-BiLSTM, R-BERT, and MTB, applied to eight subsets of the SemEval-2010 Task 8 dataset. The findings reveal a consistent pattern across all classifiers and performance metrics: increasing the dataset size initially leads to significant performance gains; however, these improvements diminish once the dataset size surpasses a certain threshold. This phenomenon of diminishing returns emphasizes the importance of balancing dataset size and quality to achieve cost-effective model training. Furthermore, the study highlights the universal nature of this pattern across different architectures and evaluation metrics, providing valuable insights for optimizing data annotation efforts in natural language processing tasks.

REFERENCES

- [1] M. Sordo and Q. Zeng-Treitler, 'On Sample Size and Classification Accuracy: A Performance Comparison', 11 2005, vol. 3745, pp. 193–201.
- [2] J. D. Prusa, T. M. Khoshgoftaar, and N. Seliya, 'The Effect of Dataset Size on Training Tweet Sentiment Classifiers', 2015 IEEE 14th International Conference on Machine Learning and Applications (ICMLA), pp. 96–102, 2015.
- [3] H. Xu, 'Discovering Knowledge in Data: An Introduction to Data Mining', Journal of Statistical Software, vol. 16, 06 2006.
- [4] R. Floca, 'Challenges of Open Data in Medical Research', 12 2014, pp. 297–307.
- [5] Isun Chehreh, Farzaneh Saadati, Ebrahim Ansari, and Bahram Sadeghi Bigham, 'Enhanced Multi-Label Question Tagging on Stack Overflow: A Two-Stage Clustering and DeBERTa-Based Approach', Proceedings of the 36th Conference of Open Innovations Association FRUCT, Helsinki, Finland, 2024, ISSN 2305-7254.
- [6] J. Shawe-Taylor, M. Anthony, and N. L. Biggs, 'Bounding sample size with the Vapnik-Chervonenkis dimension', Discrete Applied Mathematics, vol. 42, no. 1, pp. 65–73, 1993.
- [7] R. Andonie, 'Extreme Data Mining: Inference from Small Datasets', Int.J. of Computers, vol. V, pp. 280–291, 09 2010.
- [8] A. B. Dris, N. Alzakari, and H. A. Kurdi, 'A Systematic Approach to Identify an Appropriate Classifier for Limited-Sized Data Sets', 2019 International Symposium on Networks, Computers and Communications (ISNCC), pp. 1–6, 2019.
- [9] M. S. Rahman and M. Sultana, 'Performance of Firth and logF-type penalized methods in risk prediction for small or sparse binary data', BMC Med Res Methodol, vol. 17, no. 1, p. 33, Feb. 2017.
- [10] R. Andonie and L. Sasu, 'Fuzzy ARTMAP with input relevances', IEEE
- [11] transactions on neural networks / a publication of the IEEE Neural Networks Council, vol. 17, pp. 929–941, 08 2006.
- [12] D.-C. Li, W.-K. Lin, L.-S. Lin, C.-C. Chen, and W.-T. Huang, 'The attribute-trend-similarity method to improve learning performance for small datasets', International Journal of Production Research, vol. 55, pp. 1–16, 07 2016.
- [13] J. Yang, X. Yu, Z.-Q. Xie, and J.-P. Zhang, 'A novel virtual sample generation method based on Gaussian distribution', Knowledge-Based Systems, vol. 24, no. 6, pp. 740–748, 2011.
- [14] D. C. Li and C. W. Liu, 'Extending attribute information for small data set classification', IEEE Transactions on Knowledge and Data Engineering, vol. 24, no. 3, pp. 452–464, 2012.
- [15] R. Mao, H. Zhu, L. Zhang, and A. Chen, 'A New Method to Assist Small Data Set Neural Network Learning', Sixth International Conference on Intelligent Systems Design and Applications, vol. 1, pp. 17–22, 2006.
- [16] Y. Choi and H. Lee, 'Data properties and the performance of sentiment classification for electronic commerce applications', Information Systems Frontiers, vol. 19, pp. 1–20, 10 2017.
- [17] X. Zhu, C. Vondrick, C. C. Fowlkes, and D. Ramanan, 'Do We Need More Training Data?', International Journal of Computer Vision, vol. 119, no. 1, pp. 76–92, Mar. 2015.
- [18] A. Althnani et al., 'Impact of Dataset Size on Classification Performance: An Empirical Evaluation in the Medical Domain', Applied Sciences, vol. 11, no. 2, 2021.
- [19] T. Linjordet and K. Balog, 'Impact of Training Dataset Size on Neural Answer Selection Models', arXiv [cs.LG], 2019.
- [20] I. Hendrickx et al., 'SemEval-2010 Task 8: Multi-Way Classification of Semantic Relations between Pairs of Nominals', in Proceedings of the 5th International Workshop on Semantic Evaluation, 2010, pp. 33–38.
- [21] Y. Shen and X. Huang, 'Attention-Based Convolutional Neural Network for Semantic Relation Extraction', in Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers, 2016, pp. 2526–2536.
- [22] P. Zhou et al., 'Attention-Based Bidirectional Long Short-Term Memory Networks for Relation Classification', in Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), 2016, pp. 207–212.
- [23] S. Wu and Y. He, 'Enriching pre-trained language model with entity information for relation classification', in Proceedings of the 28th ACM international conference on information and knowledge management, 2019, pp. 2361–2364.
- [24] L. B. Soares, N. FitzGerald, J. Ling, and T. Kwiatkowski, 'Matching the Blanks: Distributional Similarity for Relation Learning', arXiv, 2019.